

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2020 – 2021 (Odd Semester)

INNOVATIVE PRACTICE – Unit I

Degree, Semester & Branch: VII Semester B.E. ECE B

Course Code & Title: EC8702 Adhoc and wireless Sensor Networks

Name of the Faculty member: Dr.S.Periyanayagi

ACTIVITY

Topic: Unit I Issues in Designing Routing protocol for Adhoc wireless Networks

Learning Outcome:

- The students will be able to explain the issues related to designing routing protocol.
- The students will be able to find the nodes suffering from Hidden terminal problem.

Activity Chosen: Think Pair and share

Justification:

After explaining the concept of Hidden terminal problem in adhoc wireless networks, to understand the learning level of the student about finding the nodes suffering from hidden terminal problem this activity was given. Since the classes were taken in online mode this technique was used.

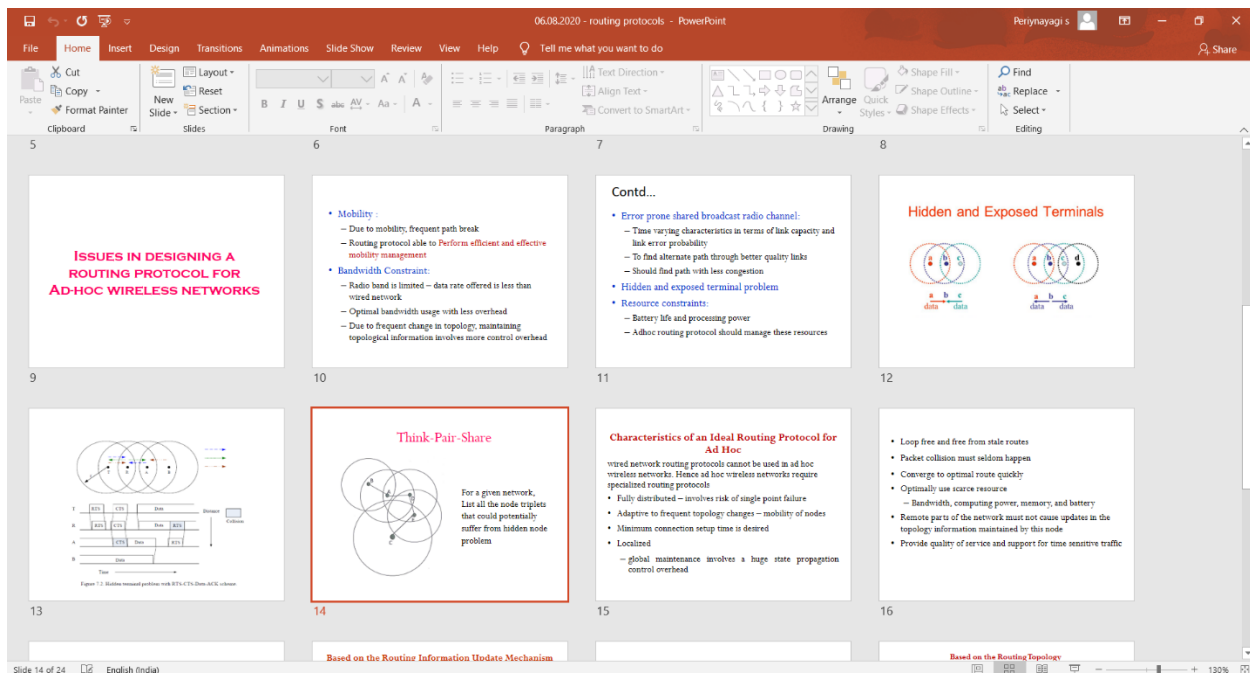
Time Allotted for the Activity: 5 Minutes

Details of the Implementation:

- Displayed the presentation slide with a case study and asked them to find the nodes after by the hidden terminal problem.
- Identified 10 pair of students and asked them to think for one minute about hidden terminal problem.
- After that, the students were asked to chat in whatsapp among themselves about hidden terminal problem.
- Finally, asked them to enter the answer in chat box.

Assessment of Effectiveness/Success of the Activity:

- ✓ This activity was effective for online mode of delivery. It helped to find the active participation of the students.
- ✓ The success of the activity is measured by the performance in the internal Assessment test and Online Quiz.



Reflective Critique:

Benefits of conducting this activity:

- Most of the students gave the right answer for the question and it helped to break the monotony during the online class.
- Interaction with the faculty member and their class mates during the online mode was effective.
- The fear of students was less compare when asking question in the online mode.

Challenges:

- Due to time constrain, 10 pair of students were made to do the activity by a random call.

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	1	1	-	-	-	-	2	-	3	-	-

CO1: The students will be able to explain the Basics of Adhoc networks and Wireless Sensor Networks

References:

1. C. Siva Ram Murthy and B. S. Manoj, —Ad Hoc Wireless Networks Architectures and Protocols, Prentice Hall, PTR, 2004.
2. An article - 'Active Learning Strategies in Face-to-Face Courses', Barbara J. Millis, The University of Texas at San Antonio Idea Paper #53, page 1-8.

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INNOVATIVE PRACTICE – Unit II

Degree, Semester & Branch: VII Semester B.E. ECE B

Course Code & Title: EC8702 Adhoc and wireless Sensor Networks

Name of the Faculty member: Dr.S.Periyanayagi

ACTIVITY

Topic: Applications of WSN

Learning Outcome:

- The students will be able to list the applications of wireless sensor networks.

Activity Chosen: One-minute paper

Justification:

The applications of Adhoc wireless networks were dealt in unit I. After explaining the concept of wireless sensor network and its architecture in order to know few applications which are familiar to the students based on real time application this activity was chosen.

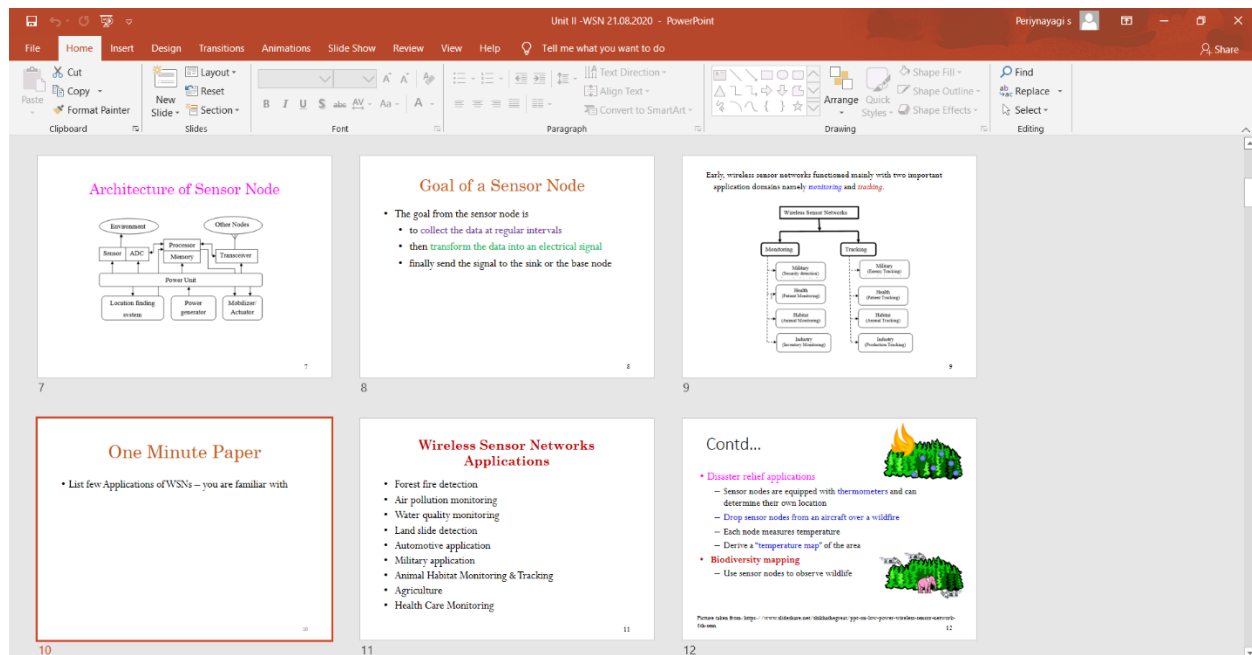
Time Allotted for the Activity: 3 Minutes

Details of the Implementation:

- The students were asked to take a paper and pen.
- They were asked to list few applications of wireless sensor networks.
- The classes were in the online mode they were asked to scan and submit the paper through LMS or email for getting attendance of that class.
- After completing the task, students uploaded the one-minute paper.

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt based on the performance of the assignment related to application of WSN.
- ✓ After completing the activity, details of application of WSN were taught to the students based on the answers given by them.



Reflective Critique:

Benefits of conducting this activity:

- The activity helped to know the involvement of the students in listening to the online class.
- It helps to avoid unwanted distractions during the online class, since the activity is time bounded.

Challenges:

- 100 % involving the students in online class were a big challenge.

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	2	-	-	-	-	1	2	-	2	-	2

CO2: The students will be able to Apply suitable routing algorithm based on network and user requirement

References:

1. Holger Karl , Andreas willig, —Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006
2. An article - ‘Active Learning Strategies in Face-to-Face Courses’, Barbara J. Millis, The University of Texas at San Antonio Idea Paper #53, page 1-8.

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INNOVATIVE PRACTICE – Unit III

Degree, Semester & Branch: VII Semester B.E. ECE B

Course Code & Title: EC8702 Adhoc and wireless Sensor Networks

Name of the Faculty member: Dr.S.Periyanayagi

ACTIVITY

Topic: MAC Protocol of WSN

Learning Outcome:

- The students will be able to explain the importance of MAC protocol in wireless sensor networks.

Activity Chosen: One-minute paper

Justification:

The concept related to MAC protocol in WSN is important. In order to make the students more attentive one-minute paper activity was planned.

Time Allotted for the Activity: 3 Minutes

Details of the Implementation:

- The students were asked to take a paper and pen.
- They were asked write the objective of MAC design of wireless sensor networks.
- The classes were in the online mode they were asked to scan and submit the paper through LMS or email for getting attendance of that class.
- After completing the task, students uploaded the one-minute paper.

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt based on the performance of internal Assessment Test 2 (Part A -3 Question and Part B 6a.) related to MAC protocol.

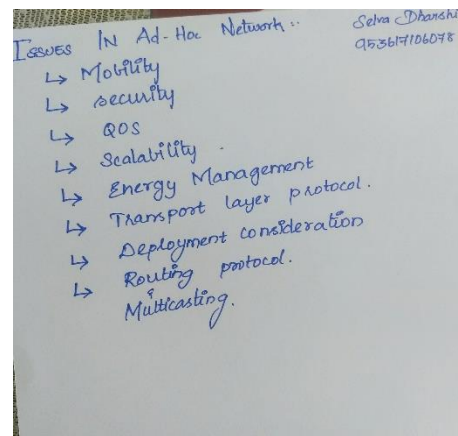
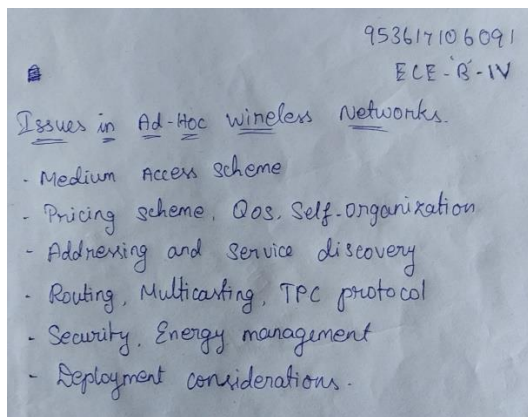
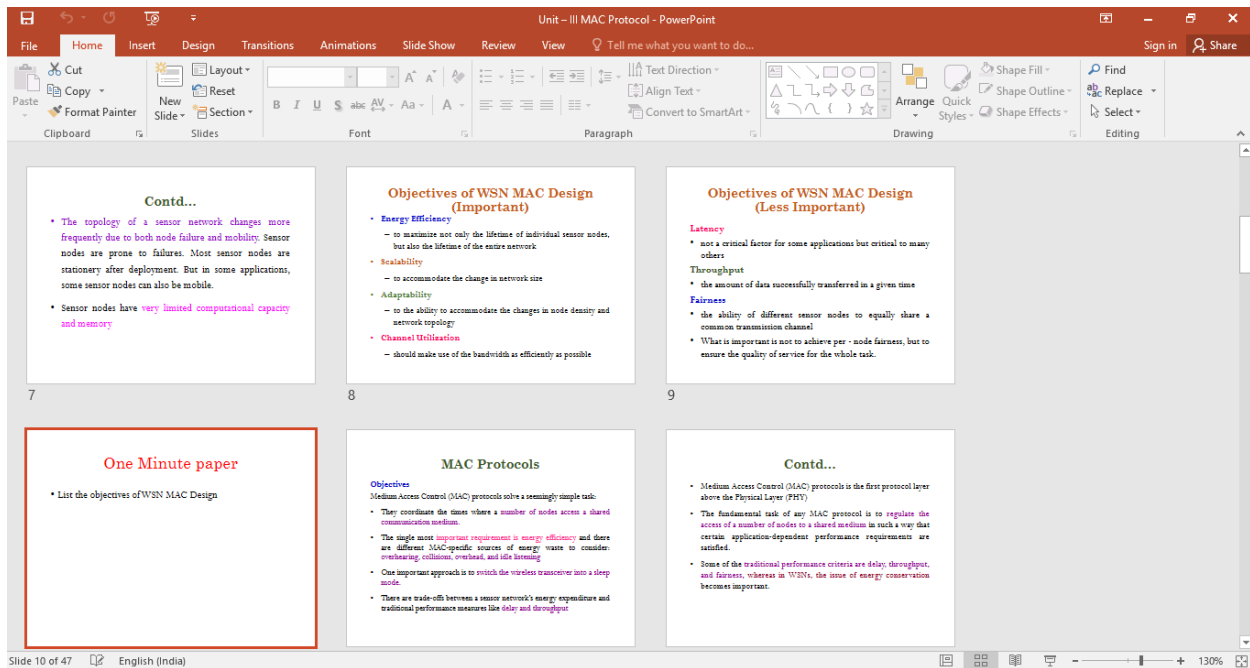
Reflective Critique:

Benefits of conducting this activity:

- The activity helped to know the involvement of the students in listening to the online class.
- It helps to avoid unwanted distractions during the online class, since the activity is time bounded.

Challenges:

- Recapping the concept in the online class was a challenging task.



Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	3	1	-	-	-	-	-	2	-	2	-	2

CO3: The students will be able to apply suitable routing algorithm based on network and user requirement

References:

1. Holger Karl , Andreas willig, —Protocol and Architecture for Wireless Sensor Networks, John wiley publication, Jan 2006
2. An article - ‘Active Learning Strategies in Face-to-Face Courses’, Barbara J. Millis, The University of Texas at San Antonio Idea Paper #53, page 1-8.